

Personalized Content Recommender System via Non-verbal Interaction Using Face Mesh and Facial Expression

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ABSTRACT

Multimedia content recommendation needs to consider users' preferences for each content. Conventional recommender systems consider them with wearable sensors, however, wearing such sensors can lead to a burden on users. In this paper, we construct a recommender system that can explicitly estimate users' preferences without wearable sensors. Specifically, by constructing lightweight but strong machine learning models suitable for our system, the users' interest levels for contents can be estimated from facial images obtained from a widely used webcam. In addition, through the interaction that the user selects displayed contents, our system finds the tendency of personal preferences for recommending contents with high user satisfaction. Our system is available on https://www.lmd-demo.org/2022/start_eng.html.

CCS CONCEPTS

• **Computer systems organization** → **Embedded and cyber-physical systems**; • **Information systems** → **Personalization**; **Recommender systems**.

KEYWORDS

Recommender system, personalization, human emotion analysis, machine learning, human-computer interaction.

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1 INTRODUCTION

Multimedia content recommendation is a fundamental task that should be personalized for the individual user [10]. The current recommender systems are personalized based on historical information (e.g., browsing history and purchase history) obtained from

each user, and the multimedia contents (e.g., images, videos, and texts) related to the obtained information are recommended for users [9]. However, since such information does not explicitly reflect the users' potential interests in multimedia contents, it can be non-beneficial information for recommendation [16]. For recommending suitable contents for the individual user from the large amounts of contents on the Web, recommender systems are desirable to fit personal preferences by introducing explicit information.

For explicitly capturing personal preferences, wearable sensors are frequently used in the recent recommender systems [3, 13, 15]. By collaboratively using the sensing information and the multimedia content information, these systems estimate the users' interest levels in the contents and make personalized recommendations related to personal preferences. However, they require users to be attached wearable sensors for perceiving personal preferences, and such requirements can be a heavy burden on users in real-world applications. Besides, since the relevance of contents previously selected by the users should be considered, previous recommender systems [3, 13, 15] do not gradually narrow down the personal preference via a step-by-step procedure that takes a lot of efforts. To effectively analyze personal preferences with less effort for users, recommender systems that can explicitly estimate personal preferences without wearable sensors are needed.

In this paper, we implement an interactive recommender system that can explicitly estimate personal preferences without wearable sensors. Different from the conventional recommender systems, our system interactively estimates personal preferences from face images captured by the widely used webcam in personal computers. Specifically, our system captures facial images from users while viewing multimedia contents, and the facial images and multimedia contents are collaboratively used for estimating personal preferences via the lightweight but strong machine learning algorithm [1, 7, 11, 14]. By using facial images, our system can alleviate the burden of wearing wearable sensors, and it leads to explicitly and easily considering personal preferences. Furthermore, to effectively perceive the tendency of individual user preferences, our system interacts non-verbally with users through the interaction that the user selects displayed contents. With these procedures, users only need to select the preferred contents that is effective for narrowing down personal preferences, and it can alleviate the efforts of the estimation.

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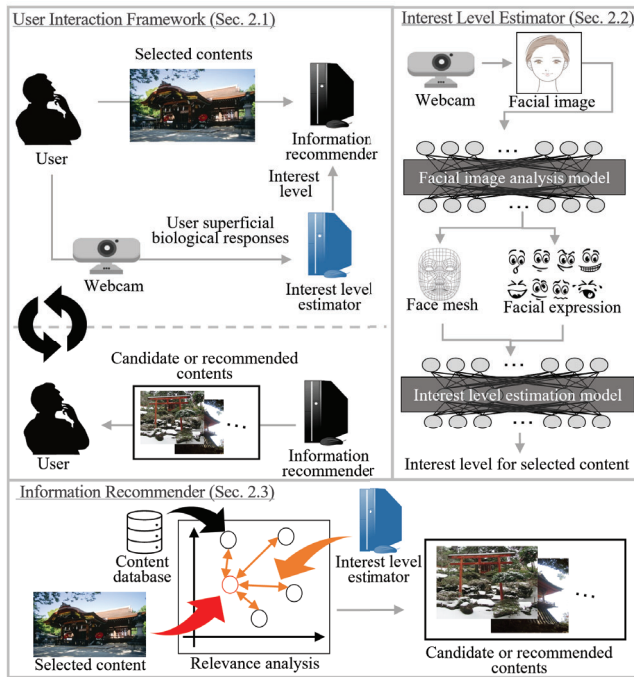


Figure 1: Overview of the proposed system. Our system interacts non-verbally with the users. Finally, our system provides personalized recommendations based on the interest levels estimated from the changes in the facial landmarks.

2 SYSTEM ARCHITECTURE

In this section, we introduce the process of the proposed demo system available on personal devices. The system overview and its user interface are respectively shown in Figs. 1 and 2.

2.1 User Interaction

For capturing the tendency of users' personal preferences, our system interacts non-verbally with users. First, the users access our system from their devices using a browser, and then, our system displays some contents. The users indicate their preferences by selecting the content that attracts their interest as their feedbacks. Besides, our system captures the users' facial image via a webcam and analyzes the relationship between the selected content and the users' preferences based on changes in the facial landmarks in facial images. Moreover, our system displays the other contents for further clarifying the users' personal preferences, and by iterating these procedures, we realize the human computer interaction for the content recommendation. Finally, after the user selects contents, our system makes personalized recommendations based on the interest level estimated from facial landmarks for the selected contents.

2.2 Interest Level Estimation

The proposed system estimates the users' interest level for the contents to analyze the personal preferences. Concretely, our system captures the users' behaviors through a webcam attached to the personal device and detects the users' face region based on the blazeface [4], which is one of the most famous real-time face detection models. Next, the face mesh and facial expressions are estimated in real time with reference to the literature [2, 8] for analyzing the temporal changes in the superficial biological responses.

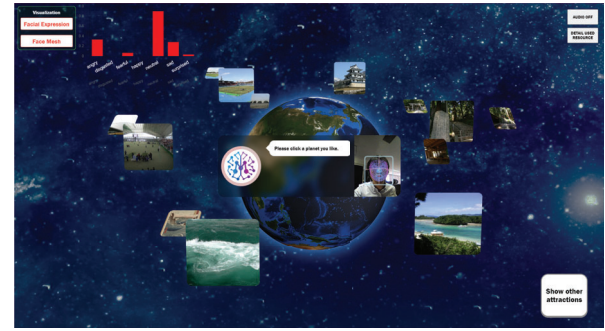


Figure 2: The user interface of our system. Our system shows several contents through the users' browser. In the upper left corner, we place buttons to visualize the elements used in the personalized recommendation.

By acquiring the superficial biological responses during the period between "the time the candidate contents are displayed" and "the time the user selects one", our system can obtain users' unconscious responses to the selected content as the non-verbal cues.

In order to estimate the interest levels for the selected content from the superficial biological responses, our system integrates the heterogeneous features obtained from facial landmarks and expressions. Concretely, we adopt the kernel multi-set canonical correlation analysis (KMCCA) [1], which can project the several input features into the common latent space, and then the extreme learning machine (ELM) [7, 11, 14] is used to estimate the interest level for the selected content from the integrated features. Here, both KMCCA and ELM have a significantly smaller number of trainable parameters than recent deep learning-based models, and we realize the personalized recommendation with these models. In this way, our system can estimate the users' interest levels to the selected contents from the superficial biological responses by applying the lightweight but strong machine learning-based method without wearing additional biomedical sensors.

2.3 Information Recommendation

Our system recommends contents tailored to individual preferences based on the interest level estimated for selected contents in Sec. 2.2. At the time of recommendation, we take the weighted average of the content-specific features calculated from selected contents by using the estimated interest levels. By calculating the cosine similarity between this average feature and the features of candidate contents, our system recommends contents similar to the selected contents. In this paper, as the content features, we adopt the visual features calculated from the content images by ResNet50 [6] pretrained on ImageNet [12] and the text features calculated from the text description of contents by WordNet [5]. In this way, we realize the interactive personalized recommendation by using the interest level estimated from the user response for the selected contents.

3 CONCLUSION

In this paper, we have presented the personalized content recommender system via non-verbal interaction using face mesh and facial expression. Our system interacts with users by allowing them to share their preferences through content selection. With the selected contents and users' superficial biological responses as a clue, our system estimates the users' interest levels in order to personalize the recommendation results.

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